Marlan and Rosemary Bourns College of Engineering

Sentry 360: 360° Security Camera

Real-Time 360° Surveillance via Distributed Edge Intelligence • CS131 • Spring 2025

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Introduction

Modern security systems fail to provide full situational awareness due to limited camera coverage and cloud dependency, creating blind spots and latency issues.

Key Problems:

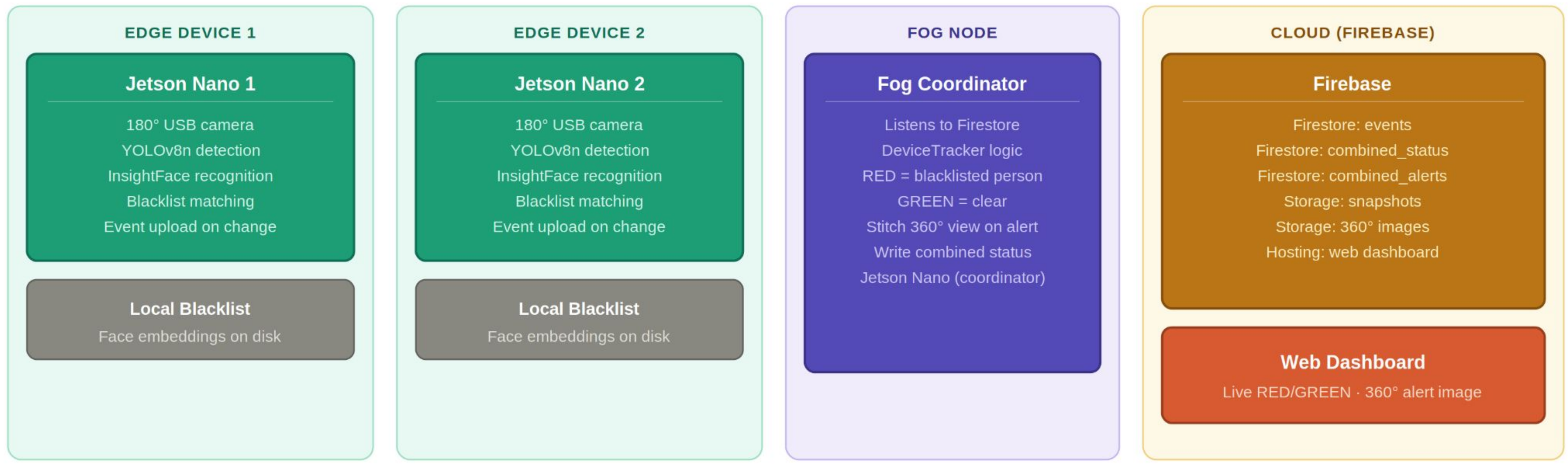
- Cloud dependency increases latency; continuous streaming wastes bandwidth
- Limited fields of view create dangerous blind spots

Design

System Architecture
Two NVIDIA Jetson Nano edge devices, each with a 180° camera, provide combined 360° coverage. YOLOv8 runs locally — only detection status is transmitted.

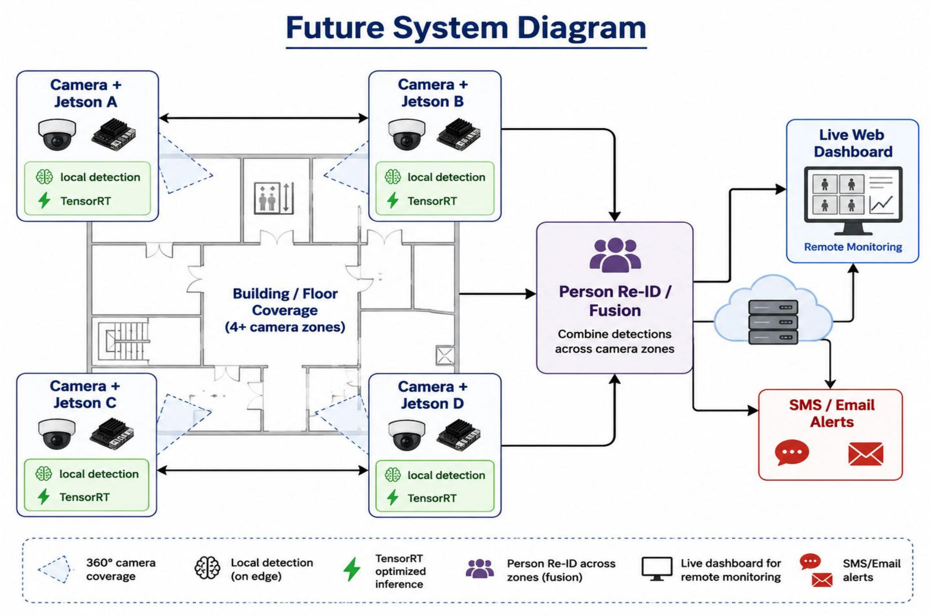
Detection Pipeline

- Edge Device 1 + 2: YOLOv8 inference → InsightFace → Firestore events → FOG decision (RED/GREEN) → Cloud logging



Future Research

- Scale to 4+ devices for full building coverage
- Add person re-ID across camera zones
- Real-time SMS/email alerts on detection
- TensorRT optimization for faster inference
- Live web dashboard for remote monitoring



Schematic

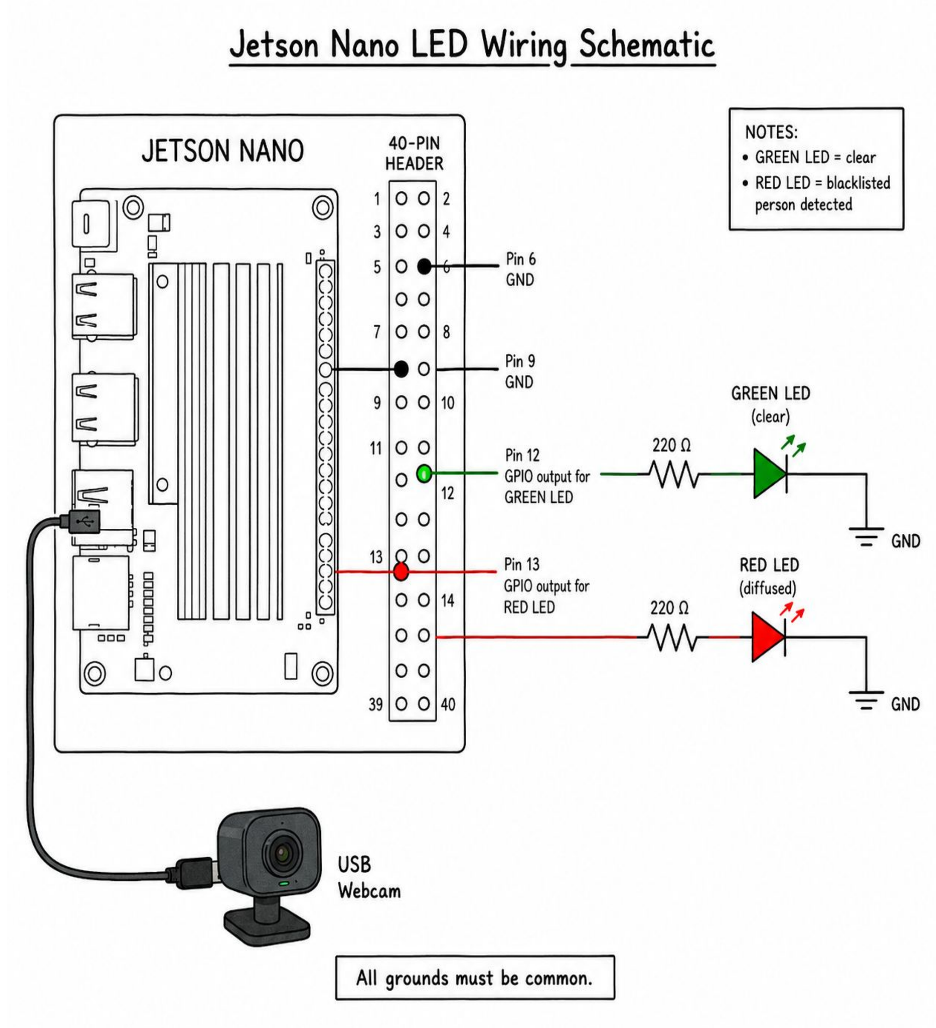
Hardware

- 2x Jetson Nano + 180° USB webcams, 3D-printed housing, LED indicator (in progress), Wi-Fi

Software

- YOLOv8, Google Firestore, InsightFace buffalo_s model, OpenCV, TensorRT (in progress)

Repo: github.com/dandyandy2004/cs131_project



Discussion

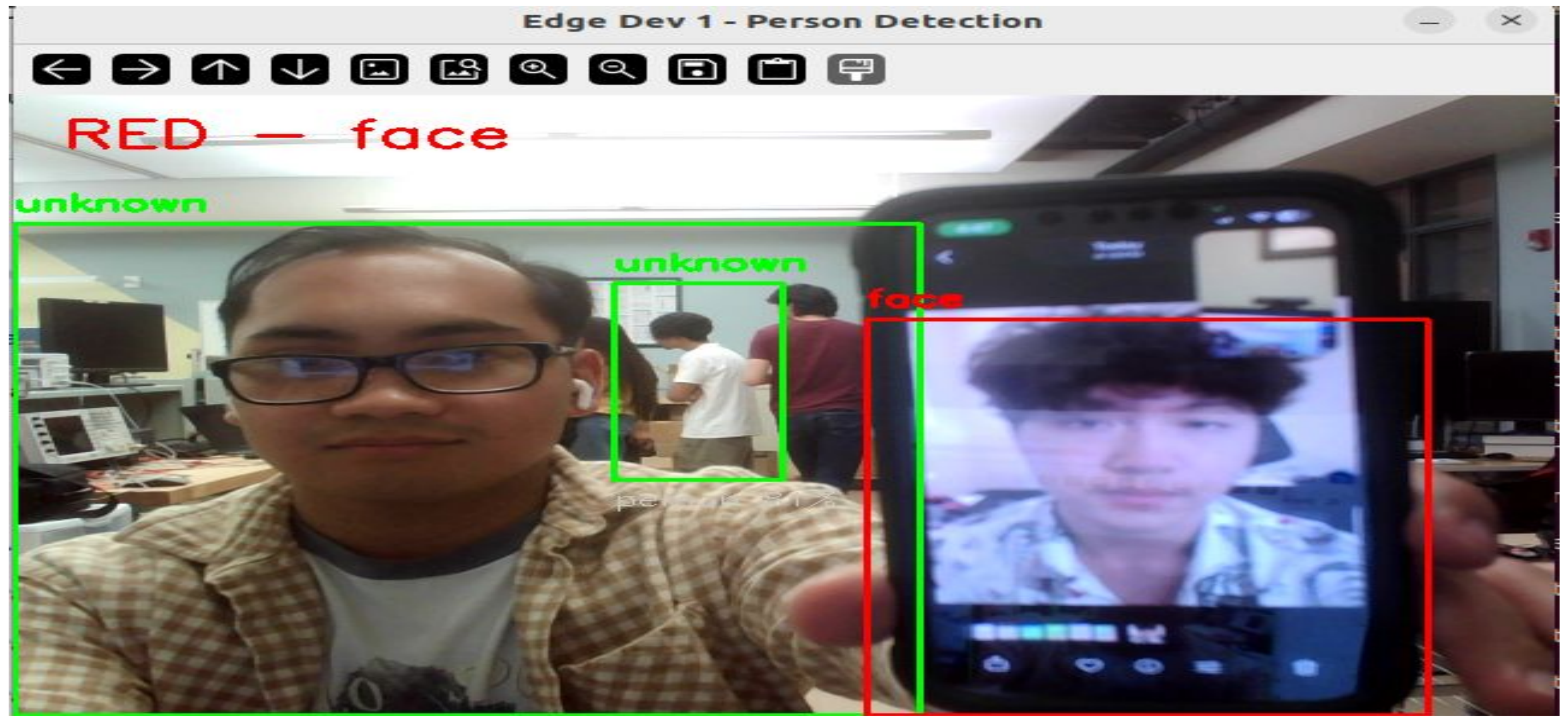
The prototype validates distributed edge intelligence: YOLOv8 inference runs fully on-device. Only alert status travels the network.

Key Achievements

- Event-driven logging reduces bandwidth vs. continuous streaming
- 3D-printed enclosure unifies Nano + webcam as one unit
- Used Firestore to log detection events in real time and uploaded image URLs for the dashboard
- Used Firebase Cloud Storage and Hosting to upload detection images and display them on a live web dashboard

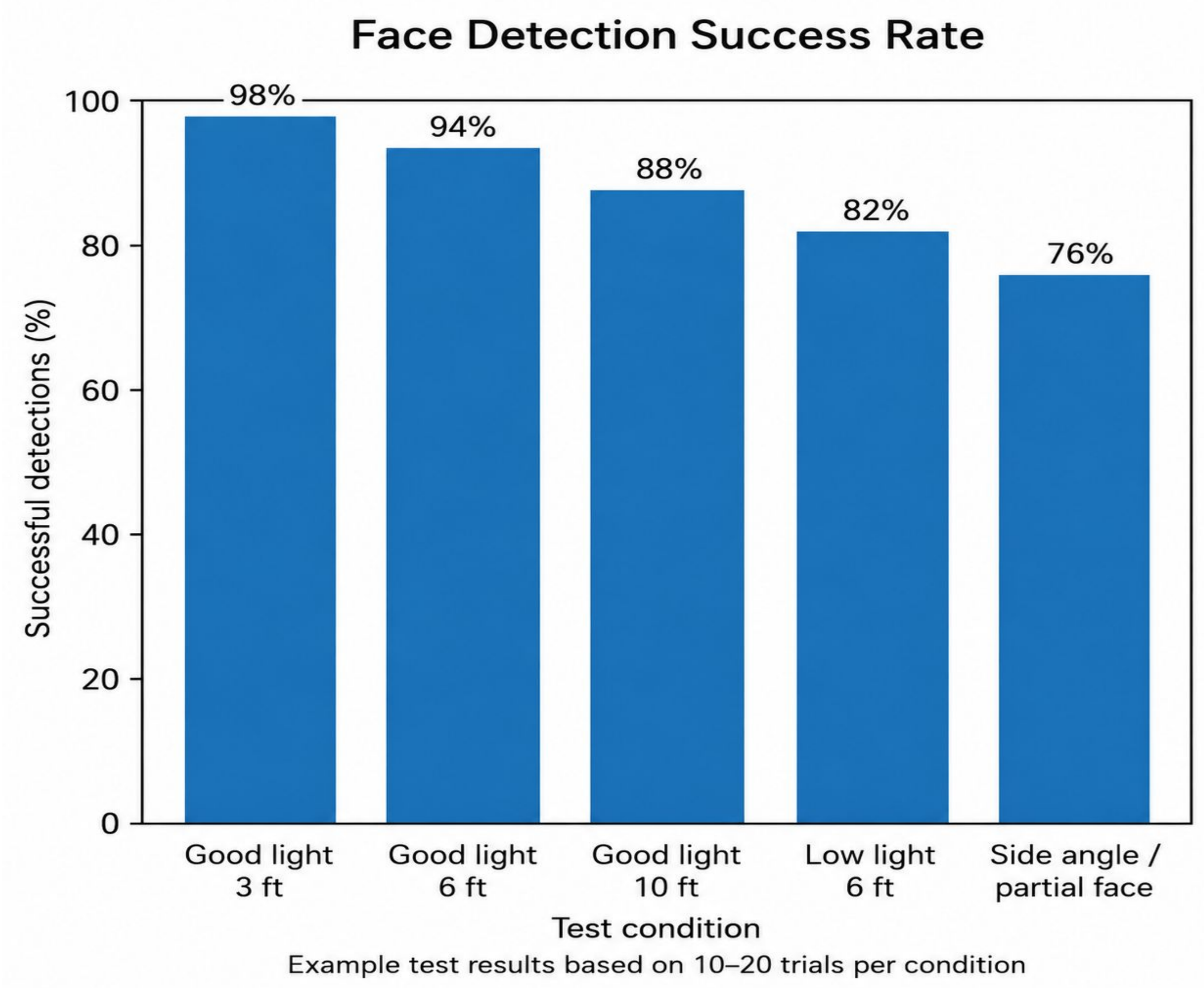
Challenges Resolved

- YOLOv8 tuned for full-body detection; USB webcam housing redesigned
- Resolved issues involving Firebase Storage permissions, Firestore configuration, and public image access to successfully connect the Jetson Nano edge devices with the cloud dashboard.



Limitations

- Nano inference speed limited without TensorRT
- Camera housing needs USB webcam fit adjustment
- 2-device prototype can't fully eliminate blind spots at scale
- Accuracy drops in low-light environments



References

- Jocher et al. (2023). YOLOv8. github.com/ultralytics
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